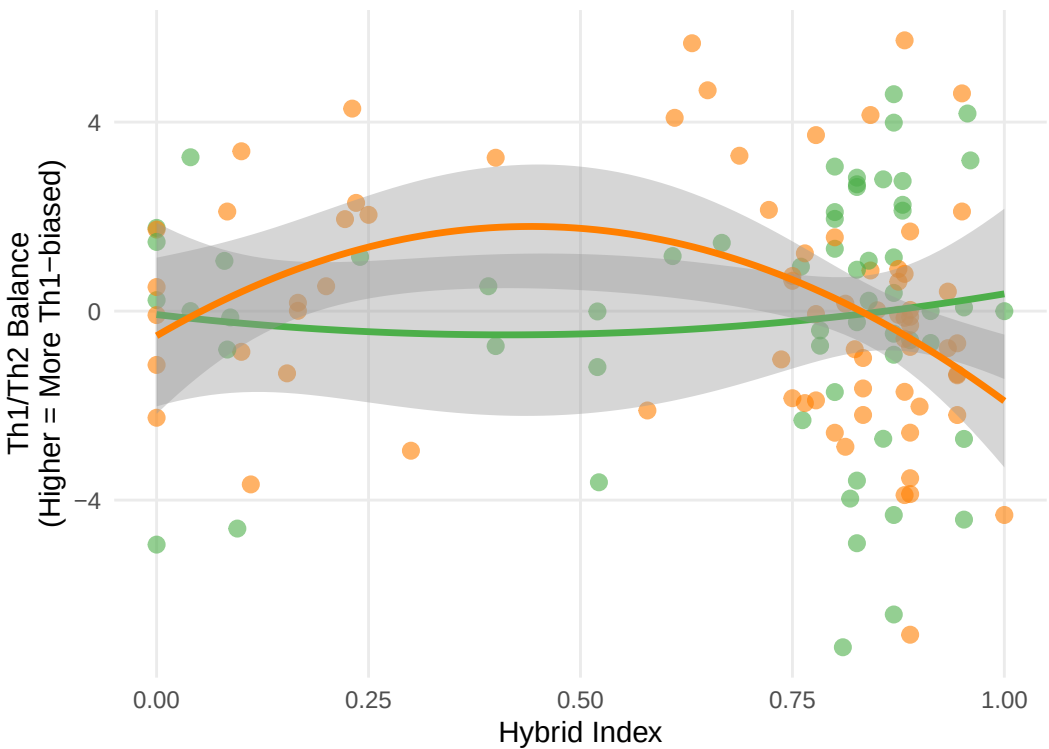


Figure 3. Sex-specific immune mechanisms underlying differential hybrid vulnerability

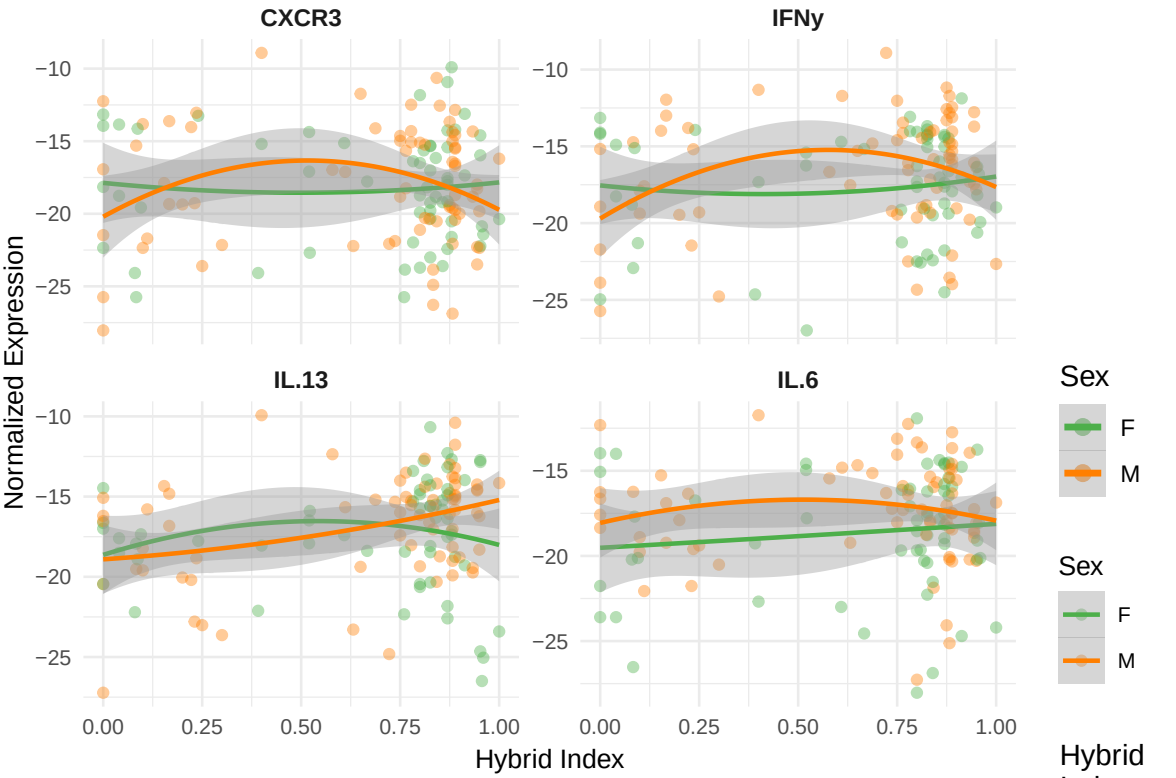
Infected mice show divergent immune trajectories with sex-specific hybrid effects

a) Th1/Th2 Balance Shows Sex-Specific Effects

Sex × He interaction in infected mice

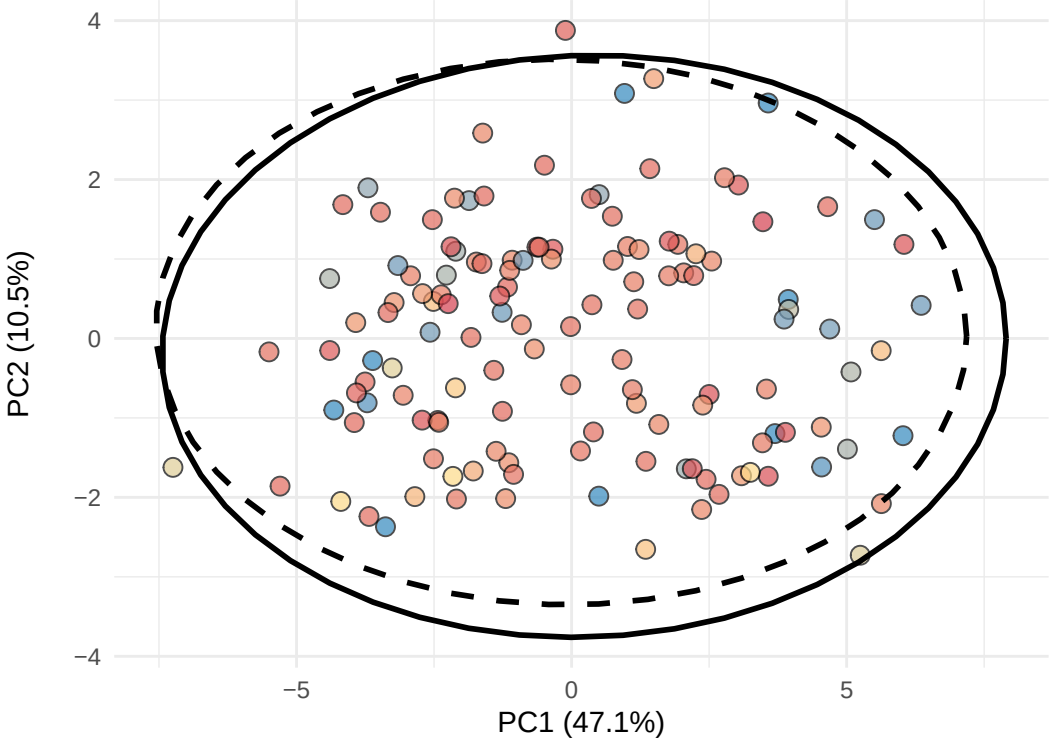


b) Key Immune Genes



c) Immune State Space Divergence

PC1 (47.1%) vs PC2 (10.5%)



d) Immune Function vs Health Cost

Th1/Th2 balance predicts weight loss

